Larkhall NDE Services Inc.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
LNDE-RPT-20160118-004-R0	Issued	2016-02-12

UT Thickness Survey - ATU-100 Full Unit

Scope and Responsibilities

Larkhall NDE Services Inc. was engaged to perform a comprehensive ultrasonic thickness survey of the ATU-100 at the Hamilton Plant. This scope encompasses a Class 2 10-year survey for lines last inspected in 2006, a Class 1 5-year survey, and specialized 3-year interval inspections for rich amine and regenerator overhead circuits. The inspection was executed in accordance with QSCP-NDT-001 Rev 2, which incorporates specific requirements for CUI program profile radiography and insulation management.

Marcus Chen served as the lead UT technician, holding SNT-TC-1A UT Level II certification. Trevor Nolan assisted in the data collection as an SNT-TC-1A UT Level I technician. The technical review and quality assurance of the collected data and final report were performed by David Park, acting as the Level III Quality Manager. All activities adhered to the personnel qualification requirements of ASNT SNT-TC-1A.

Inspection Methodology and References

Ultrasonic thickness measurements were performed using an Olympus 37DL PLUS 0611452, verified against LNDE-CAL-2016-0042. The survey followed the technical guidelines of ASTM E797 and the piping inspection requirements of API 570. For the stainless steel reflux lines 3-RF-101 and 2-RF-102, the gauge was calibrated to the specific material velocity to avoid the errors noted in HIS-2013-0451. Design parameters for the piping systems were referenced from ASME B31.3.

The inspection strategy was driven by QSCP-IP-ATU100 Rev 2, which mandates profile RT or insulation windows every 5 years for lines within the CUI temperature range. Furthermore, QSCP-DMR-2014-002 established the requirement for increased monitoring of the rich amine circuit due to higher acid-gas loading. This review moved rich amine lines to a 3-year interval and necessitated the addition of new CMLs downstream of the let-down valves.

Sour Gas and Treated Gas Circuit Results

The sour gas inlet lines 6-SG-101 and 6-SG-102 (t_nom 7.11 mm, t-min 3.56 mm) were inspected as part of the Class 1 5-year cycle. Current readings were compared against HIS-2011-0207 to determine short-term corrosion rates. Both lines remain well above their retirement thickness, with no immediate concerns regarding the design limits of 1965 kPa(g) and 66 °C.

The treated gas line 8-TG-101 (t_nom 8.18 mm, t-min 4.06 mm) was surveyed under the Class 2 10-year requirement. Data indicates stable wall thickness since the 2006 baseline. The line continues to meet the integrity requirements for its design conditions of 1896 kPa(g) and 60 °C.

Amine and Regenerator Circuit Analysis

Rich amine lines 4-RA-101, 4-RA-102, and 6-RA-103 were inspected on the 3-year interval mandated by the recent MOC. These results were compared to LNDE-RPT-20140902-024-R0 and HIS-2011-0207. The rich side exchanger piping 4-RA-104 was also monitored for accelerated thinning. Lean amine lines 4-LA-101, 4-LA-102, 6-LA-103, and 4-LA-104 were inspected under the standard Class 2 10-year cycle, showing significantly lower corrosion rates than the rich amine counterparts.

Regenerator overhead lines 3-RV-101 and 2-RV-102 were surveyed per the 3-year interval requirement. Despite the high design temperature of 121 °C, the measured thicknesses remain above the t-min values of 3.56 mm and 2.79 mm respectively. The reflux lines 3-RF-101 and 2-RF-102 were confirmed to have adequate wall thickness, utilizing the corrected velocity settings for A312 TP316 material.

Utility and Auxiliary Systems

Utility piping including reboiler steam line 2-ST-101 and cooling water lines 3-CW-101 and 3-CW-102 were inspected as Class 3 systems. The steam line 2-ST-101, with a design temperature of 186 °C, showed no significant internal degradation. The cooling water lines 3-CW-101 and 3-CW-102 remain within acceptable limits relative to their 2.79 mm t-min.

Final remaining life calculations were performed for all circuits. In accordance with site rules, the next inspection dates have been set at half the remaining life or the class maximum, whichever is sooner. The rich amine and regenerator overhead circuits remain on the accelerated 3-year schedule as per the current damage mechanism review.

Thickness readings, measured 2016-02-12

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2011-02-18	6.39	6.17	0.048	0.044	>50	2021-02-12	
6-SG-101-02	7.11	3.56	2011-02-18	6.14	5.76	0.065	0.076	28.8	2021-02-12	
6-SG-101-03	7.11	3.56	2011-02-18	6.17	5.87	0.061	0.060	38.1	2021-02-12	
6-SG-101-04	7.11	3.56	2011-02-18	6.05	5.58	0.076	0.094	21.4	2021-02-12	
6-SG-101-05	7.11	3.56	2011-02-18	5.87	5.38	0.088	0.098	18.5	2021-02-12	
6-SG-101-06	7.11	3.56	2011-02-18	6.27	5.88	0.060	0.078	29.6	2021-02-12	
6-SG-101-08	7.11	3.56	2011-02-18	6.33	6.00	0.066	0.066	36.8	2021-02-12	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2011-02-18	5.75	5.42	0.086	0.066	21.6	2021-02-12	
6-SG-102-02	7.11	3.56	2011-02-18	5.80	5.43	0.084	0.074	22.2	2021-02-12	
6-SG-102-03	7.11	3.56	2011-02-18	5.80	5.22	0.091	0.116	14.3	2021-02-12	
6-SG-102-04	7.11	3.56	2011-02-18	5.73	5.18	0.091	0.110	14.7	2021-02-12	
6-SG-102-05	7.11	3.56	2011-02-18	6.46	6.19	0.047	0.054	48.5	2021-02-12	
6-SG-102-06	7.11	3.56	2011-02-18	6.41	6.06	0.070	0.070	35.6	2021-02-12	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	2006-02-24	7.89	7.76	0.022	0.013	>50	2026-02-12	
8-TG-101-02	8.18	4.06	2006-02-24	7.75	7.46	0.037	0.029	>50	2026-02-12	
8-TG-101-03	8.18	4.06	2006-02-24	7.79	7.54	0.035	0.025	>50	2026-02-12	
8-TG-101-04	8.18	4.06	2006-02-24	7.76	7.51	0.030	0.025	>50	2026-02-12	
8-TG-101-05	8.18	4.06	2006-02-24	7.95	7.73	0.025	0.022	>50	2026-02-12	
8-TG-101-07	8.18	4.06	—	—	7.39	—	—	—	2026-02-12	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	2006-02-24	5.51	5.02	0.049	0.049	37.4	2026-02-12	
4-LA-101-02	6.02	3.18	2006-02-24	5.76	5.45	0.028	0.031	>50	2026-02-12	
4-LA-101-03	6.02	3.18	2006-02-24	5.76	5.45	0.026	0.031	>50	2026-02-12	
4-LA-101-04	6.02	3.18	2006-02-24	5.78	5.48	0.030	0.030	>50	2026-02-12	
4-LA-101-05	6.02	3.18	2006-02-24	5.81	5.54	0.024	0.027	>50	2026-02-12	
4-LA-101-06	6.02	3.18	2006-02-24	5.64	5.30	0.038	0.034	>50	2026-02-12	
4-LA-101-07	6.02	3.18	2006-02-24	5.57	4.87	0.056	0.070	24.1	2026-02-12	
4-LA-101-09	6.02	3.18	—	—	5.00	—	—	—	2026-02-12	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	2006-02-24	5.63	5.25	0.039	0.038	>50	2026-02-12	
4-LA-102-02	6.02	3.18	2006-02-24	5.67	5.36	0.036	0.031	>50	2026-02-12	
4-LA-102-03	6.02	3.18	2006-02-24	5.55	5.28	0.040	0.027	>50	2026-02-12	
4-LA-102-04	6.02	3.18	2006-02-24	5.34	4.90	0.055	0.044	31.5	2026-02-12	
4-LA-102-05	6.02	3.18	2006-02-24	5.62	5.34	0.033	0.028	>50	2026-02-12	
4-LA-102-06	6.02	3.18	2006-02-24	5.83	5.60	0.020	0.023	>50	2026-02-12	
4-LA-102-07	6.02	3.18	2006-02-24	5.63	5.40	0.030	0.023	>50	2026-02-12	
4-LA-102-08	6.02	3.18	—	—	5.70	—	—	—	2026-02-12	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	2006-02-24	6.75	6.44	0.029	0.031	>50	2026-02-12	
6-LA-103-02	7.11	3.56	2006-02-24	6.87	6.53	0.033	0.034	>50	2026-02-12	
6-LA-103-03	7.11	3.56	2006-02-24	6.65	6.28	0.040	0.037	>50	2026-02-12	
6-LA-103-04	7.11	3.56	2006-02-24	6.52	6.15	0.051	0.037	>50	2026-02-12	
6-LA-103-05	7.11	3.56	2006-02-24	6.49	5.98	0.061	0.051	39.9	2026-02-12	
6-LA-103-06	7.11	3.56	2006-02-24	6.42	5.82	0.067	0.060	33.6	2026-02-12	
6-LA-103-07	7.11	3.56	2006-02-24	6.76	6.12	0.053	0.064	39.9	2026-02-12	
6-LA-103-08	7.11	3.56	2006-02-24	6.79	6.49	0.028	0.030	>50	2026-02-12	
6-LA-103-10	7.11	3.56	—	—	6.30	—	—	—	2026-02-12	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2014-09-19	8.07	7.83	0.035	0.098	40.9	2019-02-12	
4-RA-101-02	8.56	3.81	2014-09-19	8.02	7.94	0.030	0.052	>50	2019-02-12	
4-RA-101-03	8.56	3.81	2014-09-19	7.88	7.65	0.048	0.066	>50	2019-02-12	
4-RA-101-04	8.56	3.81	2014-09-19	8.18	8.00	0.032	0.042	>50	2019-02-12	
4-RA-101-05	8.56	3.81	2014-09-19	7.98	7.82	0.039	0.068	>50	2019-02-12	
4-RA-101-06	8.56	3.81	2014-09-19	7.91	7.87	0.029	0.029	>50	2019-02-12	
4-RA-101-07	8.56	3.81	2014-09-19	8.22	8.14	0.023	0.032	>50	2019-02-12	
4-RA-101-08	8.56	3.81	2014-09-19	7.71	7.31	0.069	0.142	24.6	2019-02-12	
4-RA-101-09	8.56	3.81	2014-09-19	8.27	8.13	0.022	0.038	>50	2019-02-12	
4-RA-101-10	8.56	3.81	2014-09-19	8.21	8.02	0.033	0.052	>50	2019-02-12	
4-RA-101-11	8.56	3.81	2014-09-19	7.88	7.60	0.054	0.088	42.9	2019-02-12	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2014-09-19	8.25	8.02	0.026	0.062	>50	2019-02-12	
4-RA-102-02	8.56	3.81	2014-09-19	7.89	7.64	0.047	0.070	>50	2019-02-12	
4-RA-102-03	8.56	3.81	2014-09-19	7.95	7.55	0.051	0.098	38.0	2019-02-12	
4-RA-102-04	8.56	3.81	2014-09-19	7.93	7.72	0.041	0.048	>50	2019-02-12	
4-RA-102-05	8.56	3.81	2014-09-19	7.84	7.67	0.044	0.080	48.1	2019-02-12	
4-RA-102-06	8.56	3.81	2014-09-19	7.95	7.73	0.039	0.056	>50	2019-02-12	
4-RA-102-07	8.56	3.81	2014-09-19	7.98	7.78	0.040	0.064	>50	2019-02-12	
4-RA-102-08	8.56	3.81	2014-09-19	7.58	7.35	0.057	0.106	33.3	2019-02-12	
4-RA-102-09	8.56	3.81	2014-09-19	7.84	7.54	0.051	0.088	42.2	2019-02-12	
4-RA-102-11	8.56	3.81	2014-09-19	8.12	7.96	0.046	0.046	>50	2019-02-12	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2014-09-19	10.62	10.29	0.036	0.036	>50	2019-02-12	
6-RA-103-02	10.97	4.57	2014-09-19	10.06	9.69	0.070	0.108	47.2	2019-02-12	
6-RA-103-03	10.97	4.57	2014-09-19	10.39	10.06	0.047	0.072	>50	2019-02-12	
6-RA-103-04	10.97	4.57	2014-09-19	9.79	9.31	0.084	0.132	35.8	2019-02-12	
6-RA-103-05	10.97	4.57	2014-09-19	9.81	9.48	0.074	0.124	39.5	2019-02-12	
6-RA-103-06	10.97	4.57	2014-09-19	10.43	10.32	0.035	0.048	>50	2019-02-12	
6-RA-103-07	10.97	4.57	2014-09-19	10.37	10.20	0.040	0.070	>50	2019-02-12	
6-RA-103-08	10.97	4.57	2014-09-19	10.49	10.33	0.032	0.022	>50	2019-02-12	
6-RA-103-09	10.97	4.57	2014-09-19	10.39	10.33	0.034	0.046	>50	2019-02-12	
6-RA-103-10	10.97	4.57	2014-09-19	9.93	9.49	0.077	0.112	43.8	2019-02-12	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-11	10.97	4.57	2014-09-19	10.10	9.77	0.236	0.236	22.0	2019-02-12	
6-RA-103-13	10.97	4.57	2014-09-19	10.38	10.15	0.060	0.060	>50	2019-02-12	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2013-09-09	6.49	6.34	0.063	0.081	34.2	2019-02-12	
3-RV-101-02	7.62	3.56	2013-09-09	6.44	6.31	0.063	0.052	43.5	2019-02-12	
3-RV-101-03	7.62	3.56	2013-09-09	5.80	5.63	0.095	0.098	21.2	2019-02-12	
3-RV-101-04	7.62	3.56	2013-09-09	6.13	5.88	0.086	0.102	22.8	2019-02-12	
3-RV-101-05	7.62	3.56	2013-09-09	6.48	6.36	0.066	0.050	42.6	2019-02-12	
3-RV-101-06	7.62	3.56	2013-09-09	6.11	5.96	0.082	0.087	27.7	2019-02-12	
3-RV-101-07	7.62	3.56	2013-09-09	5.73	5.54	0.122	0.122	16.2	2019-02-12	
3-RV-101-08	7.62	3.56	2013-09-09	6.36	6.31	0.074	0.074	37.2	2019-02-12	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2013-09-09	4.49	4.35	0.059	0.037	26.4	2019-02-12	
2-RV-102-02	5.54	2.79	2013-09-09	4.29	4.01	0.079	0.109	11.2	2019-02-12	
2-RV-102-03	5.54	2.79	2013-09-09	4.68	4.64	0.042	0.037	43.9	2019-02-12	
2-RV-102-04	5.54	2.79	2013-09-09	4.81	4.73	0.043	0.020	45.5	2019-02-12	
2-RV-102-05	5.54	2.79	2013-09-09	4.96	4.97	0.013	0.013	>50	2019-02-12	
2-RV-102-06	5.54	2.79	2013-09-09	4.55	4.27	0.102	0.102	14.6	2019-02-12	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2013-12-09	5.37	5.31	0.009	0.017	>50	2026-02-12	
3-RF-101-02	5.49	2.41	2013-12-09	5.40	5.34	0.006	0.003	>50	2026-02-12	
3-RF-101-03	5.49	2.41	2013-12-09	5.37	5.33	0.006	0.012	>50	2026-02-12	
3-RF-101-04	5.49	2.41	2013-12-09	5.41	5.35	0.006	-0.002	>50	2026-02-12	
3-RF-101-05	5.49	2.41	2013-12-09	5.32	5.33	0.006	0.007	>50	2026-02-12	
3-RF-101-06	5.49	2.41	2013-12-09	5.17	5.23	0.015	0.017	>50	2026-02-12	
3-RF-101-07	5.49	2.41	2013-12-09	5.48	5.37	0.003	-0.002	>50	2026-02-12	
3-RF-101-09	5.49	2.41	2013-12-09	5.26	5.19	-0.029	-0.029	>50	2026-02-12	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2013-12-09	3.60	3.76	0.008	0.009	>50	2026-02-12	
2-RF-102-02	3.91	2.16	2013-12-09	3.70	3.64	0.012	0.024	>50	2026-02-12	
2-RF-102-03	3.91	2.16	2013-12-09	3.79	3.75	0.009	0.005	>50	2026-02-12	
2-RF-102-04	3.91	2.16	2013-12-09	3.82	3.38	0.025	0.043	28.3	2026-02-12	
2-RF-102-05	3.91	2.16	2013-12-09	3.67	3.49	0.082	0.082	16.1	2024-03-06	
2-RF-102-06	3.91	2.16	2013-12-09	3.67	3.47	0.099	0.099	13.2	2022-09-26	
2-RF-102-07	3.91	2.16	2013-12-09	3.72	3.57	0.020	0.028	>50	2026-02-12	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	2006-02-24	5.75	5.31	0.038	0.044	48.2	2026-02-12	
4-LA-104-02	6.02	3.18	2006-02-24	5.56	5.00	0.052	0.056	32.4	2026-02-12	
4-LA-104-03	6.02	3.18	2006-02-24	5.74	5.42	0.027	0.032	>50	2026-02-12	
4-LA-104-04	6.02	3.18	2006-02-24	5.86	5.63	0.022	0.023	>50	2026-02-12	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-05	6.02	3.18	2006-02-24	5.51	4.88	0.058	0.063	26.9	2026-02-12	
4-LA-104-06	6.02	3.18	2006-02-24	5.83	5.55	0.028	0.028	>50	2026-02-12	
4-LA-104-07	6.02	3.18	2006-02-24	5.62	5.28	0.037	0.034	>50	2026-02-12	
4-LA-104-08	6.02	3.18	2006-02-24	5.50	5.11	0.042	0.039	46.3	2026-02-12	
4-LA-104-10	6.02	3.18	—	—	4.84	—	—	—	2026-02-12	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2014-09-19	7.84	7.71	0.040	0.048	>50	2019-02-12	
4-RA-104-02	8.56	3.81	2014-09-19	8.09	7.85	0.035	0.058	>50	2019-02-12	
4-RA-104-03	8.56	3.81	2014-09-19	8.46	8.28	0.057	0.056	>50	2019-02-12	
4-RA-104-04	8.56	3.81	2014-09-19	8.49	8.18	0.063	0.084	>50	2019-02-12	
4-RA-104-05	8.56	3.81	2014-09-19	8.37	8.16	0.050	0.082	>50	2019-02-12	
4-RA-104-06	8.56	3.81	2014-09-19	7.91	7.61	0.049	0.074	>50	2019-02-12	
4-RA-104-07	8.56	3.81	2014-09-19	7.61	7.32	0.064	0.092	38.0	2019-02-12	
4-RA-104-08	8.56	3.81	2014-09-19	8.00	7.81	0.074	0.074	>50	2019-02-12	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	2006-02-24	3.64	3.61	0.016	0.003	>50	2026-02-12	
2-ST-101-02	3.91	2.29	2006-02-24	3.67	3.55	0.018	0.012	>50	2026-02-12	
2-ST-101-03	3.91	2.29	2006-02-24	3.82	3.55	0.020	0.027	46.5	2026-02-12	
2-ST-101-04	3.91	2.29	2006-02-24	3.77	3.68	0.007	0.009	>50	2026-02-12	
2-ST-101-05	3.91	2.29	2006-02-24	3.72	3.50	0.021	0.022	>50	2026-02-12	
2-ST-101-06	3.91	2.29	—	—	3.63	—	—	—	2026-02-12	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	2006-02-24	5.13	5.03	0.020	0.010	>50	2026-02-12	
3-CW-101-02	5.49	2.79	2006-02-24	5.20	5.01	0.020	0.019	>50	2026-02-12	
3-CW-101-03	5.49	2.79	2006-02-24	5.08	4.93	0.030	0.015	>50	2026-02-12	
3-CW-101-04	5.49	2.79	2006-02-24	5.38	5.26	0.010	0.012	>50	2026-02-12	
3-CW-101-05	5.49	2.79	2006-02-24	5.33	5.35	0.007	-0.002	>50	2026-02-12	
3-CW-101-06	5.49	2.79	—	—	5.08	—	—	—	2026-02-12	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	2006-02-24	5.20	5.06	0.022	0.014	>50	2026-02-12	
3-CW-102-02	5.49	2.79	2006-02-24	5.02	4.37	0.057	0.065	24.2	2026-02-12	
3-CW-102-03	5.49	2.79	2006-02-24	5.07	4.54	0.050	0.053	32.9	2026-02-12	
3-CW-102-04	5.49	2.79	2006-02-24	5.03	4.69	0.041	0.034	46.2	2026-02-12	
3-CW-102-05	5.49	2.79	—	—	4.66	—	—	—	2026-02-12	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT Level II; CAN/CGSB-48.9712 UT Level II	2016-02-12	_____
Prepared	Trevor Nolan	UT technician, SNT-TC-1A UT Level I	2016-02-12	_____
Reviewed	David Park	Level III / Quality Manager, SNT-TC-1A UT Level III	2016-02-12	_____